## Comparative Analysis

Modelling approaches and their relevance to this project

| Approach | Strengths | Weaknesses | Relevance to this project |
|---|---|---|---|
| **Logistic Regression / scorecard** | Transparent coefficients; easy benchmarking; familiar governance; relatively simple calibration | May miss nonlinear interactions; requires careful transformations and feature design | Primary interpretable benchmark |
| **Random Forest / gradient boosting** | Captures nonlinear effects and interactions; often strong tabular performance | More difficult to explain directly; can require calibration and tuning; may be unstable under drift | Candidate nonlinear comparison |
| **Naive academic ML benchmark** | Easy to implement; low software cost; useful for teaching | Often uses random splits, leakage-prone preprocessing, immature labels, and accuracy-focused metrics | The revised workflow is designed to correct these weaknesses |
| **Commercial AI platforms** | Integrated deployment, monitoring, governance, and experimentation tools | Licensing cost, vendor dependence, restricted transparency, and implementation complexity | Industry comparator, not a direct numerical benchmark |
| **Internal institutional scorecard** | Access to proprietary data, policies, protected-group monitoring, and operational outcomes | High development, governance, and data-integration requirements | More realistic production reference |
| **Proposed framework** | Reproducible open-source workflow; historical cohort selection; chronological validation; separate outcome table; explainability-ready architecture | Not production validated; no live data; no verified protected-group data; LGD/EAD not yet empirically modelled | Academic proof of concept and methodological framework |

*Table 2.1: Comparative Analysis of Credit Risk Frameworks and Commercial Platforms*